

From Narrative Inflation to Verifiable Self-Correction: An Empirical Longitudinal Case Study of Multi-Turn LLM Mathematical Reasoning on the Riemann Hypothesis

從敘事膨脹到可驗證自我修正：大語言模型在長程多輪前沿數學推理中的失敗模式
與修正機制實證研究

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AI Systems Involved & Roles:

- **Google Gemini** (Gemini 3.7 Flash/Pro) — Reasoning Co-Pilot / Proposer
- **Perplexity** (active session) — Formal adversarial red-team auditor for Prüfer/Dirac track (Entries 303–385 / Rounds 103–146)
- **OpenAI ChatGPT** — Independent adversarial reviewer for LaTeX proof-draft track (Weil positivity / Suzuki screw-kernel)
- **Perplexity** (search-mode session) — Retrieval-grounded literature research assistant (Connes–Consani–Moscovici line)

August 2026 • Dataset & Repository: <https://github.com/chienhaoc/riemann-hypothesis>

⚠ **Author Action Required Before Submission:**

1. **Model identity:** Confirm literal model versions. Cross-referencing shows journal entries 303–385 correspond to the Perplexity session (Rounds 103–146), where "ChatGPT" was used as an informal placeholder, whereas the `paper.tex` review transcript is a genuine OpenAI ChatGPT conversation.
2. **Total counts (388 entries / 145 review cycles):** Entry-pair 387–388 is verified in `journal/2026-08-14.md` ("388" is grounded). Reconcile exact maximum count of "Review N" occurrences before final camera-ready submission.

Abstract—Current benchmarks for evaluating mathematical reasoning in Large Language Models (LLMs)—GSM8K, MATH, OlympiadBench—focus on static, single-turn, closed-form contest problems with known ground truths, and so cannot capture how frontier systems behave across hundreds of iterative turns on a genuinely unsolved research problem.

We present a longitudinal case study spanning **388 chronological research-progression entries** in a human–AI collaborative attempt to reduce the Riemann Hypothesis (RH) to a spectral/operator-theoretic statement via a symplectic Dirac-operator construction, cross-checked against an independently generated **10-class taxonomy of reasoning failure modes**, each grounded in verbatim transcript excerpts and independent symbolic (SymPy) re-derivation. We additionally document an **eleventh failure mode—citation overreach**, discovered in a parallel, mathematically distinct proof attempt (a Weil-positivity / de Branges route) that was reviewed independently: the model cited real, verifiable arXiv papers (Groskin 2026, Suzuki 2026) whose authors explicitly flag the needed convergence/positivity statement as an *open problem*, yet the model asserted the statement as an established input to its own "proof."

The most striking single artifact we recovered is a sequence of entries (251–258) in which the model issued an explicit, unqualified **"100% Grand Seal—Riemann Hypothesis proven"** claim, later retracted under adversarial symbolic counter-proof. Across at least seven independently verifiable multi-turn chains, precise, quantitative, symbolically-checkable counter-proofs reliably induced the model to abandon a flawed claim and produce a corrected, independently re-verified derivation within one to two turns. We release the full transcript dataset for independent audit.

1. Introduction and Methodology

1.1 Why Long-Horizon, Open-Problem Case Studies Matter

Standard benchmarks miss three critical dynamics: (i) they have known ground truth, so they cannot show what a model does when *nobody* knows the answer; (ii) they are single-turn, missing error accumulation across extended dialogues; (iii) without execution/verification tools, models generate fluent pseudo-proofs that are difficult to distinguish from genuine progress.

1.2 The Actual Multi-Session, Multi-Model Architecture

Role	System	Artifact Reviewed	Approach
Reasoning Co-Pilot / Proposer	Gemini (3.7 Flash/Pro)	journal/2026-08-14.md , entries 1–388	Symplectic Dirac operator / Prüfer dynamics / $S(X, t)$
Red-Team Auditor (this session)	Perplexity	Entries $\approx$ 303–385 (Rounds 103–146)	Same track, independent adversarial review
Independent Reviewer (separate thread)	OpenAI ChatGPT	paper.tex , paper(1).tex	Weil quadratic-form positivity, de Branges, Suzuki screw-kernel
Literature Research Assistant	Perplexity (search-mode)	Connes–Consani–Moscovici line	Retrieval-grounded literature survey

1.3 Canonical Indexing

To keep the taxonomy auditable, we use native entry-pair numbering from journal/2026-08-14.md , mapped to human-facing round numbers via the empirically confirmed relation:

$$\text{entry} = 2 \times \text{round} + 93$$

(verified at rounds 105, 107, 112, 126, 127, 130–132, 136–139, 144–145 against entries 303, 307, 317, 345, 347, 353–357, 365–373, 381–383 respectively).

2. Taxonomy of Eleven Long-Horizon Reasoning Failure Modes

Mode 1: Scale and Coordinate Dimension Confusion

Evidence (Entry 319–320 $\rightarrow$ corrected 321–322; Round 112 $\rightarrow$ 113): The model substituted $X = t$ into $\phi_0(X, t) = \frac{t}{2}X$, yielding $\mathcal{O}(t^2)$ growth versus the correct $\vartheta(t) \sim \frac{t}{2}\log(t/2\pi e) \in \mathcal{O}(t \log t)$. The correction in Entry 321–322 redefined the substitution on the logarithmic coordinate $X_t = \log(t/2\pi e)$ and reproduced $\vartheta(t)$ with zero discrepancy.

Mode 2: Hidden Circular Reasoning

Evidence (Entry 345–346, Round 126): The model asserted an unconditional decay $e^{-X/2}$ by assuming zeros lie on the critical line ($\text{Re}(\rho) = 1/2$), contradicting its own unconditional Vinogradov–Korobov bound $\mathcal{O}_t(e^{-cX^{1/3}})$. Corrected in Entry 347–348 by strictly labeling conditional and unconditional tracks.

Mode 3: Category Mixing between Ensemble Statistics and Pointwise Bounds

Evidence (Entries 351–359, Rounds 128–132): Conflating $\langle \text{Re } \mathcal{C}_2 \rangle \equiv 0$ with pointwise $|S(X, t_0)| \leq \mathcal{O}_{t_0}(X)$. Verified via SymPy:

```
import sympy as sp
X, t, T, Ct = sp.symbols('X t T C_t', positive=True)
F = sp.Rational(1,2)*X**2*t # leading part of int_0^t |S(X,u)|^2 du
integral = T**2*F.subs(t,T) - sp.integrate(2*t*F, (t,0,T)) # = X^2*T^3/6
avg_ReC2 = -sp.Rational(1,8*T)*integral + (X**2/(16*T))*sp.integrate(t**2,(t,0,T))
assert sp.simplify(avg_ReC2) == 0 # confirms exact 0*X^2*T^2 cancellation
```

Mode 4: Topological Confusion of Isolated vs. Essential Accumulation

Evidence (Entries 381–384, Rounds 144–145): Arguing that $\det_3 \rightarrow 0$ on a shrinking band I_X contradicts $\sigma_{\text{ess}} = \emptyset$. Retracted in Entry 383–384 after noting $\text{Leb}(I_X) \sim e^{(\beta_0-1)X} \rightarrow 0$ collapses to an isolated point eigenvalue, which is completely compatible with discrete spectrum.

Mode 5: Weight Mismatch in Formula Transplantation

Evidence (Entry 323–324, Round 114–115): Equating $-\zeta'/\zeta$ (weight $\log p/\sqrt{p}$) directly with Selberg's $S(T)$ expansion (weight $1/\sqrt{p}$), corrected via explicit Abel summation by parts.

Mode 6: Heavy-Machinery Invocation for Elementary Facts

Evidence (Entry 303–304, Round 105): Invoking Baker's theorem on linear forms in logarithms for $\mathbb{Q}$ -linear independence of $\{\log p\}$, which follows trivially from unique prime factorization.

Mode 7: Narrative Progress Inflation & The "100% Grand Seal"

Flagship Evidence (Entries 251–258): Labeled Theorem 251.3 (Grand Seal), Theorem 253.2 (Grand Seal), and Theorem 255.2 (Grand Seal), asserting an unqualified proof of RH, with Entry 257–258 synthesizing a "Blueprint" to merge Tiers 1–3B. Retracted under adversarial refutation.

Mode 8: Unchecked Perturbation-Expansion Validity Domains

Evidence (Entries 371–374, Rounds 139–141): Expanding $\sqrt{1 - 4W^2/X^4}$ around 0 where $4W^2/X^4 \approx 1/4 \rightarrow 0$. Corrected by isolating the radical non-perturbatively.

Mode 9: Notation-Masked Unproven Assumptions

Evidence (Entries 279–280, 355–356, Rounds 130–131): Gauge masking of divergences, resolved by deriving the unique $\mathfrak{sl}(2, \mathbb{R})$ Lie generator from first principles.

Mode 10: Adversarially Induced True Self-Correction

Documented Chains: Entries 319→321, 345→347, 351→353→355, 371→373→375, 381→383. Seven independently re-verified chains in this session.

Mode 11 (New): Citation Overreach

Evidence (ChatGPT review of paper.tex): Citing Groskin (2026) and Suzuki (2026) as establishing convergence/spectral results that those authors explicitly labeled as open conjectures. This mode cannot be caught by symbolic computation alone, requiring external literature retrieval.

3. Discussion

3.1 Rhetorical Framing: Gamified framing correlated with earlier inflation; percentage language was suppressed following explicit policy intervention.

3.2 Retrieval Grounding: Search-mode sessions with retrieval grounding exhibited markedly lower rates of overreaching claims than open-ended generative proof attempts.

3.3 Epistemic Limitations: All auditors in this pipeline are LLMs; symbolic checks certify local algebra but cannot certify formal global correctness.

3.4 Scope Correction Relative to Earlier Drafts: Acknowledging that earlier paper drafts also exhibited inadvertent numerical precision inflation, which was caught and corrected through the same discipline recommended by the paper.

4. Conclusion and Data Release

The complete transcript dataset, cross-reference index, and this paper are available at:
<https://github.com/chienhaoc/riemann-hypothesis> .

References

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